

2080

Bachelor of Science in Computer Science and Information Technology

Course Title: **Cryptography**

Code No: **CSC327** Semester: **5th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Define discrete logarithms. How key generation, encryption and decryption is done in RSA. In a RSA cryptosystem, given $p=13$ and $q=7$, determine private key, public key, and perform encryption and decryption for the text $M= "hi"$ using 0 to 25 for letters from a to z.
2. Write down the encryption and decryption process at 2-DES and 3-DES. Explain the Feistel cipher structure. Divide $5x^2 + 4x + 6$ by $2x + 1$ over GF (7).
3. What are the applications of hash functions? Discuss how SHA-1 algorithm generates hash value from a given message.

Section B

Attempt any Eight questions[8×5=40]

4. Show encryption and decryption of "csit" using hill cipher having key $K = 3257$
5. Give the formal definition of Authentication system. Describe about one way and mutual Authentication system.
6. List the stage of certificate life cycle. What are the types of firewalls?
7. What is malicious logic? How Zombies are different from Trojan horses?
8. How Miller Rabin test is used for primality testing? Show whether the number 561 passes the test.
9. Show that the set of integers is Ring under addition and multiplication.
10. How substitution ciphers are different from transposition ciphers? Given a message $M= "CSIT PROGRAM IS A HOT CAKE"$, encrypt M using Rail Fence cipher with rail size=3.
11. Describe about IPSec. List the five services of PGP.
12. How does meet in middle attack work in Diffie Hellman key Exchange protocol? Explain.